

ONLINE FOOD ORDERING SYSTEM (YUM-YUM)

A PROJECT REPORT

Submitted by

MR. SOHAIM ASLAM AZMI

MR. AYAN AHMAD

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INTEGRAL UNIVERSITY



CERTIFICATE

Certified that this project report **“ONLINE FOOD ORDERING SYSTEM (YUM-YUM)”** is the bonafide work of **“MR. SOHAIM ASLAM AZMI & MR. AYAN AHMAD”** who carried out the project work under my supervision.

Mr. Khalid Jamal

Associate Professor

Department of Computer Application

Integral University, Lucknow



INTEGRAL UNIVERSITY



CERTIFICATE

Certified that this project report **“ONLINE FOOD ORDERING SYSTEM (YUM-YUM)”** is the bonafide work of **“MR. SOHAIM ASLAM AZMI & MR. AYAN AHMAD”** who have successfully carried out the project.

Dr. Md. Faizan Farooqui

Project Coordinator

Department of Computer Application

Integral University, Lucknow

Dr. Mohd. Faisal

Head

Department of Computer Application

Integral University, Lucknow

DECLARATION

“I hereby declare that this submission is my own work and that, to the best of my knowledge and belief, it contains no material previously published or written by another person nor material which has been accepted for the award of any other degree or diploma of the university or other institute of higher learning, except where due acknowledgment has been made in the text”.

Date: 6/6/2024

Student Sign:

Mr. Sohaim Aslam Azmi

Student Sign:

Mr. Ayan Ahmad

Acknowledgement

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MR. AYAN AHMAD

TABLE OF CONTENTS

CHAPTER NO.	TITLE	PAGE NO.
	ABSTRACT	x
	LIST OF TABLES	xi
	LIST OF FIGURES	xi
	LIST OF ABBREVIATIONS	xii
1.	INTRODUCTION	1
	1.1. OBJECTIVE	2
	1.2. MODULES	3
	1.2.1. Administrator Module	3
	1.2.2. Customer Module	3
	1.2.3. Admin Module	3
	1.2.4. User Module	4
2.	HARDWARE AND SOFTWARE	
	REQUIREMENTS	6
	2.1. SOFTWARE REQUIREMENTS	6
	2.2. HARDWARE REQUIREMENTS	6
3.	PROBLEM IDENTIFICATION AND	
	FEASIBILITY STUDY	7
	3.1. PROBLEM IDENTIFICATION	7
	3.2. FEASEIBILITY STUDY	8
4.	REQUIREMENT ANALYSIS	11

5.	REVIEW OF THE PREVIOUS WORK	13
6.	PROJECT DESCRIPTION	15
	6.1. LITERATURE SURVEY	15
	6.1.1. Aim	15
	6.1.2. Existing System	16
	6.1.3. Proposed System	16
	6.2. FRONT-END	17
	6.2.1. HTML	17
	6.2.2. JavaScript	18
	6.2.3. CSS	19
	6.3. BACK-END	20
	6.3.1.PHP	20
	6.4. DATABASE MANAGEMENT	21
	6.4.1. MySQL	21
	6.5. PURPOSE	22
	6.6. SYSTEM ARCHITECTURE	24
	6.7. METHODOLOGY	26
	6.8. DEVELOPMENT TOOLS	28

	6.9. TESTING	31
	6.10. SYSTEM LIMITATIONS	33
	6.11. SYSTEM STRENGTHS	33
7.	DESIGN	34
	7.1. Introduction	34
	7.2. UML Diagrams	34
	7.3. Use Case	34
	7.4. USE CASE DIAGRAMS	36
	7.4.1. Admin Use Case Diagram	37
	7.4.2. User Use Case Diagram	38
	7.5. DATA FLOW DIAGRAM	39
	7.6. E-R DIAGRAM	42
	7.7. FLOWCHART	45
	7.8. DATABASE DESIGN	46
	7.8.1. Database Tables	46
	7.9. SNAPSHOTS OF THE PROJECT	52
	7.9.1. Home Page	52
	7.9.2. Customer Registration Page	52

	7.9.3. Menu Page	53
	7.9.4. Contact Us Page	54
	7.9.5. Admin Panel	54
8.	CONCLUSION	56
9.	FUTURE WORK	57
10.	REFERENCES	58
11.	BIODATA OF EACH GROUP MEMBER	59
	11.1. MR. SOHAIM ASLAM AZMI	59
	11.2. MR. AYAN AHMAD	61

ABSTRACT

An Online Food Ordering System is proposed here which simplifies the food ordering process. The proposed system shows a user interface and update the menu with all available options so that it eases the customer work. Customer can choose more than one item to make an order and can view order details before logging off. The order confirmation is sent to the customer. The order is placed in the queue and updated in the database and returned in real time. This system assists the staff to go through the orders in real time and process it efficiently with minimal errors. Online food ordering system is mainly designed primarily function for use in the food delivery industry. In the modern food industries allows to quickly and easily delivery on customer place. Restaurant employees then use these orders through an easy to delivery on customer place easy find out navigate graphical interface for efficient processing.

Our proposed system is an online food ordering system that enables ease for the customers. It overcomes the disadvantages of the traditional queueing system. Our proposed system is a medium to order online food hassle free from restaurants as well as mess service. This system improves the method of taking the order from customer. The online food ordering system sets up a food menu online and customers can easily place the order as per their wish. Also, with a food menu, customers can easily track the orders. This system also provides a feedback system in which user can rate the food items. Also, the proposed system can recommend hotels, food, based on the ratings given by the user, the hotel staff will be informed for the improvements along with the quality. The payment can be made online or pay-on-delivery system. For more secured ordering separate accounts are maintained for each user by providing them an ID and a password.

LIST OF TABLES

1. Admin Table
2. Cart Table
3. Messages Table
4. Orders Table
5. Products Table
6. Users Table

LIST OF FIGURES

Figure 1: System Architecture of Online Food Ordering System

Figure 2: Diagram for Admin Use Case

Figure 3: Diagram for User Use Case

Figure 4: DFD Diagram for Online Food Ordering System

Figure 5: E-R Diagram for Online Food Ordering System

Figure 6: Flowchart Diagram for Food Ordering System

Figure 7: Database Design for Food Ordering System

Figure 8: Database Table Schema

Figure 9: Website Homepage

Figure 10: User Registration Page

Figure 11: Food Menu Page

Figure 12: Contact Us Page

Figure 13: Admin Panel Page

LIST OF ABBREVIATIONS

HTML: HyperText Markup Language

PHP: Hypertext Preprocessor (originally stood for Personal Home Page)

JS: JavaScript

CSS: Cascading Style Sheets

MySQL: My Structured Query Language

XAMPP: Cross-Platform (X), Apache, MySQL, PHP, Perl

DFD: Data Flow Diagram

ER: Entity-Relationship

UML: Unified Modelling Language

OS: Operating System

IDE: Integrated Development Environment

1. INTRODUCTION

The online food ordering system sets up a food menu online and customers can easily place the order as per they like. Also, with a food menu, online customers can easily track the orders. The management maintains customers database, and improve food delivery service. The Restaurant management systems motivates us to develop the system. There are various facilities provided so that the users of the system will get service effectively. Also, the system considers Restaurants as well as Mess facility to the customers. Again, the idea comes that mostly mess users are person who are shifted for various reason in new cities. So, they are interrelated. Increasing use of smart phones is also considered as a motivation, so that any users of this system get all service on single click. Another motivation can be considered as the system will be designed to avoid users doing fatal errors, users can change their own profile, users can track their food items through GPS, users can provide feedback and recommendations and can provide reviews.

In recent years, the advent of technology has transformed many aspects of our lives, including how we order and experience food. One of the most significant advancements in the food industry is the emergence of online food ordering systems. These platforms have revolutionized the way people order meals, offering unparalleled convenience, accessibility, and efficiency.

In the following sections, we will provide a comprehensive overview of the theoretical foundations, technical components, experimental results, and practical implications of customer services, employee services, ordering and delivery services, aiming to offer valuable insights and inspirations for developers and researchers.

1.1. OBJECTIVE

The main objective of this project is to develop an application which gives provision to the restaurant owners to flourish their business by uploading menus at no cost and will invariably lead to high customer retention and acquisition rates.

- 1. Reliability:** Build a reliable and robust system that can handle high traffic volumes, process orders efficiently, and maintain uptime during peak hours. Implement measures to prevent system failures, errors, and delays that could impact the user experience.
- 2. Security:** Prioritize the security of user data, payment information, and transactions. Implement encryption, authentication, and secure payment gateways to protect sensitive information and prevent unauthorized access or fraud.
- 3. Customer Satisfaction:** Focus on delivering exceptional customer service and satisfaction. Provide responsive customer support channels, handle inquiries and complaints promptly, and address issues related to orders, deliveries, or payments effectively.
- 4. Very Less Paper Work:** Online Food Ordering System requires less paper works. In this project all record is fetched directly into the computer and reports can be generated through just a click to save time.
- 5. Personalization:** Offer personalized recommendations and promotions based on user preferences, order history, and location. Allow users to save favorite orders, customize preferences, and receive notifications about special offers and discounts.

1.2. MODULES

- Administrator Module
- Customer Module

1.2.1. Administrator Module:

Admin can view all the information about the user to edit all the details about the customer.

- Create food category
- Manage food categories
- Add food item
- Manage food item
- Manage user order

1.2.2. Customer Module:

These functionalities are provided:

- View product's list
- Register
- Place orders

1.2.3. Admin Module:

- 1. Restaurant Management:** Admins can oversee restaurant profiles by adding, editing, or removing details like name, location, contact information, operating hours, and menu items.

2. **Menu Management:** Admins have control over restaurant menus, allowing them to add new dishes, update prices, descriptions, and availability, and categorize items for ease of navigation.
3. **Order Management:** Admins can efficiently handle orders, including tracking order status, managing order updates, and processing refunds or cancellations when required.
4. **User Management:** Admins can manage user accounts effectively, including tasks such as registering new users, updating profiles, resetting passwords, and adjusting permissions for staff members.
5. **Analytics and Reporting:** Admins gain insights into various metrics such as order volumes, sales revenue, popular menu items, customer feedback, and delivery times through detailed reports, aiding in informed decision-making.
6. **Search Order:** In this section, admin can search for particular order with the help of order number.
7. **Reports:** In this section, the admin can view order details, order counts and sales report according to dates. Admin can also update his profile, change passwords, recover passwords and accounts.

1.2.4. User Module:

1. **Account Registration and Profile Management:** Users can create accounts and manage personal details, including contact information and delivery addresses, to customize their experience and preferences.

- 2. Browsing and Ordering:** Users can explore restaurant menus, view dish details, and place orders for delivery or pickup, with options for customization and adding special instructions.
- 3. Search and Filter:** Users can search for restaurants and menu items based on criteria like cuisine type, location, and price range, using filter options to refine their choices for easier decision-making.
- 4. Cart Management:** Users can add items to their shopping carts, adjust quantities, and review orders before checkout, ensuring accuracy and completeness of their selections.
- 5. Checkout and Payment:** Users are guided through secure checkout processes, choosing payment options and providing delivery details, with transactions handled securely to complete orders.
- 6. Order Tracking and History:** Users can monitor the status of their orders in real-time, receiving updates on preparation, delivery, and estimated arrival times, and access a history of past orders for reference.
- 7. Feedback and Reviews:** Users can provide feedback and reviews for restaurants and menu items, sharing their experiences to help others make informed decisions and provide valuable insights for restaurant improvement.

2. HARDWARE AND SOFTWARE REQUIREMENTS

2.1. SOFTWARE REQUIREMENTS:

Windows 10 or higher

Languages Used:

Front-End: HTML, JavaScript, CSS

Back-End: PHP3.

Database Management: MySQL

Operating System: The script should be compatible with various operating systems like Windows, macOS, and Linux, as long as there is stable browser with internet connection.

Softwares Used:

Web Servers: XAMPP, WAMPP

IDEs: Visual Studio Code, Notepad ++

Browser: Chrome, Firefox

2.2. HARDWARE REQUIREMENTS:

Processor: Any Intel Core i3 or AMD Ryzen 3

Processor Speed: 2.60 GHz

Graphics Card: AMD Radeon

Ram: 1 GB (minimum)

Hard Disk: 10 GB (minimum) HDD/SSD

Platform: Windows 7/8/10/11 64-Bit

3. PROBLEM IDENTIFICATION AND FEASIBILITY STUDY

3.1. PROBLEM IDENTIFICATION:

1. **Security Concerns:** Users may be hesitant to provide personal information or payment details due to concerns about data privacy, security breaches, or fraudulent activities.
2. **Delivery Issues:** Late deliveries, missing items, or poor packaging quality can lead to negative reviews and damage the reputation of the system and its partner restaurants.
3. **Limited Restaurant Options:** A limited selection of restaurants or cuisines may reduce user satisfaction and limit the system's appeal to a wider audience.
4. **Lack of Integration:** Poor integration with third-party services such as delivery partners, payment gateways, or inventory management systems can lead to inefficiencies and errors in order processing.
5. **Scalability Challenges:** Difficulty in handling increased traffic, orders, or user accounts during peak times can lead to system slowdowns or failures, impacting user experience and revenue generation.
6. **Limited Features:** The system may lack essential features such as order tracking, user reviews, personalized recommendations, or integration with popular payment methods, limiting its functionality and usability.
7. **Inaccurate Information:** Incorrect or outdated menu items, prices, or restaurant details can lead to misunderstandings, order errors, and customer dissatisfaction.

- 8. Poor Communication:** Lack of communication between users and customer support, delayed responses to inquiries, or insufficient updates on order status can result in dissatisfaction and loss of trust.

3.2. FEASIBILITY STUDY:

1. Technical Feasibility:

- **Available Technology:** Evaluate the existing technology stack and infrastructure to determine if it can support the development and operation of the system. Consider factors such as web development frameworks, databases, hosting platforms, and scalability requirements.
- **Development Tools:** Assess the availability and suitability of development tools and resources needed to build the system, including programming languages, integrated development environments (IDEs), version control systems, and third-party APIs.
- **Expertise:** Evaluate the technical expertise and skills required to implement and maintain the system. Determine if in-house resources possess the necessary knowledge or if external expertise is needed to fill any gaps.

2. Operational Feasibility:

- **Staff Training:** Assess the readiness of staff members to adopt and use the online food ordering system effectively. Determine if training programs or support materials will be required to familiarize users with the system's features and functionality.

- **Workflow Integration:** Evaluate how the system will integrate with existing operational workflows and processes within restaurants, delivery services, and customer support teams. Identify any potential bottlenecks or challenges that may arise during implementation.
- **Disruption Mitigation:** Develop strategies to minimize disruptions to existing processes and ensure a smooth transition to the new system. Consider phased implementation approaches, pilot testing, and contingency plans to address unforeseen issues.

3. Financial Feasibility:

- **Development Costs:** Estimate the initial investment required to develop the online food ordering system, including expenses for software development, hardware infrastructure, and licensing fees.
- **Ongoing Maintenance Expenses:** Calculate the recurring costs associated with maintaining and supporting the system, including server hosting, software updates, bug fixes, and technical support.
- **Revenue Streams:** Identify potential sources of revenue, such as transaction fees, subscription models, advertising revenue, or partnerships with restaurants and delivery services. Evaluate the projected return on investment (ROI) and profitability of the system over time.

4. Market Feasibility:

- **Demand Analysis:** Conduct market research to assess the demand for online food ordering services within the target demographic. Analyze

consumer preferences, behavior patterns, and willingness to adopt digital platforms for ordering food.

- **Competitor Analysis:** Evaluate the competitive landscape by analyzing existing online food ordering platforms, their features, pricing models, and market share. Identify opportunities to differentiate the system and attract users.
- **Trend Analysis:** Monitor emerging trends in the food delivery industry, such as changes in consumer preferences, technology adoption, and regulatory developments. Anticipate future market dynamics and adapt the system accordingly.

5. Legal and Regulatory Feasibility:

- **Data Privacy Laws:** Ensure compliance with data privacy regulations, such as the General Data Protection Regulation (GDPR) in Europe or the California Consumer Privacy Act (CCPA) in the United States. Implement measures to protect user data and obtain necessary consent for data processing.
- **Food Safety Regulations:** Adhere to food safety standards and regulations imposed by local health authorities or government agencies. Ensure that restaurants and delivery partners comply with food handling, storage, and sanitation requirements.
- **Intellectual Property Rights:** Respect intellectual property rights related to trademarks, copyrights, and patents when developing and branding the online food ordering system. Obtain necessary licenses or permissions for using third-party content or software components.

4. REQUIREMENT ANALYSIS

1. Functional Requirements:

- **User Registration and Authentication:** Users should be able to register for an account on the platform and authenticate themselves securely to access personalized features.
- **Menu Browsing:** Users should be able to browse through menus of different restaurants, view dish details, prices, and images, and filter or search for specific items.
- **Order Placement:** Users should be able to select items from the menu, add them to their cart, specify quantity and customization options, and place orders for delivery or pickup.
- **Payment Processing:** The system should support various payment methods such as credit/debit cards, digital wallets, or cash on delivery, and ensure secure payment processing.
- **Order Tracking:** Users should be able to track the status of their orders in real-time, from preparation to delivery, and receive notifications or updates on order progress.
- **Feedback and Reviews:** Users should be able to provide feedback and reviews for restaurants and menu items, share their experiences, and rate the quality of service.
- **Admin Management:** Administrators should have access to tools for managing restaurant listings, menus, orders, user accounts, and other system settings.

- **Reporting and Analytics:** The system should generate reports and analytics on key metrics such as order volumes, sales revenue, customer feedback, and delivery performance to help stakeholders make informed decisions.

2. Non-Functional Requirements:

- **Performance:** The system should be responsive and scalable to handle concurrent user traffic, ensuring fast loading times and minimal downtime during peak hours.
- **Usability:** The user interface should be intuitive, easy to navigate, and accessible across different devices and screen sizes, catering to users with varying levels of technical proficiency.
- **Security:** The system should implement robust security measures to protect user data, payment information, and system integrity, including encryption, authentication, and authorization mechanisms.
- **Reliability:** The system should be reliable and resilient, with built-in redundancy and failover mechanisms to minimize service disruptions and data loss.
- **Scalability:** The system should be designed to scale horizontally or vertically to accommodate growth in user base and transaction volume without sacrificing performance or reliability.
- **Availability:** The system should be available and accessible to users 24/7, with minimal planned downtime for maintenance or upgrades, and rapid recovery from unplanned outages.

5. REVIEW OF PREVIOUS WORK

Previous work in the field of real-time color detection has laid the groundwork for the development of systems like the one described in this project.

1. Existing Techniques:

Explore the range of existing techniques employed in online food ordering systems, including web-based platforms, mobile applications, cloud-based solutions, and integration with emerging technologies like AI, IoT, and blockchain.

2. Hardware and Software Platforms:

Examine the hardware and software platforms commonly used in online food ordering systems, such as servers, databases, web frameworks (e.g., Django, Ruby on Rails), mobile operating systems (e.g., iOS, Android), and development tools (e.g., IDEs, version control systems).

3. Applications and Use Cases:

Analyze the various applications and use cases of online food ordering systems, including consumer-facing platforms for browsing menus, placing orders, and tracking deliveries, as well as backend systems for restaurant management, order processing, and delivery logistics.

4. Challenges and Limitations:

Identify the challenges and limitations faced by online food ordering systems, such as competition from established players, customer retention, delivery logistics, food quality control, regulatory

compliance, cybersecurity risks, and scalability issues during peak demand.

5. Performance Evaluation:

Evaluate the performance of online food ordering systems based on key metrics such as response times, uptime, order accuracy, delivery speed, customer satisfaction scores, and business metrics like revenue growth, market share, and customer retention rates.

6. Future Directions:

Discuss potential future directions for research and development in online food ordering systems, including the adoption of emerging technologies like AI-driven personalization, chatbots for customer support, blockchain for supply chain transparency, and sustainability initiatives to reduce food waste and environmental impact.

6. PROJECT DESCRIPTION

The Online Food Ordering System is a web-based platform designed to streamline the process of ordering food from various restaurants, providing a seamless experience for users to browse menus, place orders, and get food delivered to their doorstep. It includes features such as user registration, restaurant and menu management, search and browse capabilities, order placement, secure payment integration, real-time delivery tracking, notifications, and reviews. Built with a robust architecture using modern frontend technologies like HTML, CSS, JavaScript, backend technologies like PHP, and databases such as MySQL, it ensures data security, scalability, and a user-friendly interface. The implementation involves detailed requirement analysis, design, parallel development, rigorous testing, and deployment on cloud platforms, followed by ongoing maintenance. This system addresses potential challenges like data privacy, traffic management, and third-party integrations, aiming to enhance customer satisfaction and operational efficiency for restaurants and delivery personnel.

6.1. LITERATURE SURVEY

6.1.1. Aim:

The aim of the literature survey is to explore existing research and systems related to online food ordering, identify their strengths and weaknesses, and propose enhancements or novel approaches to improve the efficiency, user experience, and functionality of online food ordering systems. Primary aims of the literature survey is to shed lights on the common vulnerabilities and challenges and threats of the application system.

6.1.2. Existing System:

In the existing system, online food ordering platforms typically offer basic functionalities such as browsing menus, placing orders, and making payments. These systems often lack advanced features such as personalized recommendations, real-time order tracking, and seamless integration with external services. User feedback may highlight issues with usability, order accuracy, or delivery times. Furthermore, security concerns related to data privacy and payment processing may be raised.

6.1.3. Proposed System:

The proposed system aims to address the limitations of existing systems by introducing innovative features and enhancements. This includes personalized recommendations based on user preferences and order history, real-time order tracking using GPS technology, and integration with third-party services such as food delivery aggregators and payment gateways to improve convenience and reliability. Additionally, measures to enhance security and privacy, such as data encryption and secure payment processing, will be implemented to ensure user trust and confidence in the system. Overall, the proposed system aims to provide a more seamless, user-friendly, and secure online food ordering experience for customers.

6.2. FRONT-END

6.2.1. HTML:

HTML, or Hypertext Markup Language, is the standard markup language used to create and design web pages. It provides the basic structure and content of a webpage by using a system of tags and elements. HTML allows developers to define the layout, format text, embed images, create links, and add multimedia content such as audio and video. It serves as the foundation of web development, providing the structure upon which websites are built and enabling users to access and interact with information on the internet.

1. **Creating Links:** HTML allows developers to create hyperlinks that connect different pages within the website. For example, links can be used to navigate between restaurant listings, menu pages, and the checkout process.
2. **Embedding Images:** HTML provides tags for embedding images into web pages. In an online food ordering system, images of dishes, restaurant logos, and promotional banners can be displayed using HTML `img` tags to enhance visual appeal and attract users.
3. **Displaying Text Content:** HTML is used to display text content such as restaurant descriptions, menu items, prices, and order details. Text can be formatted using HTML tags like headings, paragraphs, lists, and emphasis tags to improve readability and presentation.
4. **Implementing Forms:** HTML forms are essential for collecting user input, such as delivery addresses, customization options, and payment details. Forms allow users to interact with the website by entering information and submitting orders securely.

6.2.2. JAVASCRIPT:

JavaScript is a versatile programming language commonly used in web development to add interactivity and dynamic functionality to web pages. It runs on the client-side, meaning it executes code within the user's web browser, enabling actions like form validation, animations, and asynchronous communication with servers.

- 1. Form Validation:** JavaScript ensures data accuracy by validating user input in forms, checking for errors such as missing fields or incorrect information, and providing instant feedback to users, preventing submission of incomplete or erroneous orders.
- 2. Dynamic Content Updates:** JavaScript updates content dynamically without page reloads, such as updating the shopping cart or displaying real-time order status updates, providing a seamless and responsive user experience.
- 3. Asynchronous Requests (AJAX):** JavaScript facilitates background communication with the server, allowing for asynchronous data retrieval without disrupting user interaction. This enables features like live search, auto-complete suggestions, and dynamic loading of restaurant listings and menu items.
- 4. User Authentication and Authorization:** JavaScript implements user authentication and authorization mechanisms, enabling registered users to log in, manage accounts, and access personalized features like order history and saved preferences, enhancing security and personalization.

6.2.3. CSS:

CSS or Cascading Style Sheets is essential for creating a visually appealing, user-friendly, and accessible online food ordering system that effectively conveys the brand's identity and provides a seamless experience for users.

- 1. Layout and Structure:** CSS defines the layout and arrangement of elements, ensuring consistency and visual appeal across the website's pages.
- 2. Typography:** CSS styles text content, improving readability and user experience by customizing fonts, sizes, and alignments.
- 3. Color Scheme and Visual Elements:** CSS creates a cohesive design by defining color schemes, backgrounds, borders, and other visual elements, reinforcing branding and identity.
- 4. Responsive Design:** CSS enables the website to adapt to different devices and screen sizes, providing a seamless experience for users on desktops, tablets, and smartphones.
- 5. Animations and Transitions:** CSS adds interactivity and visual interest through animations and transitions, enhancing user engagement and feedback during the ordering process.
- 6. Accessibility:** CSS contributes to accessibility by styling elements for readability and usability, ensuring the website is inclusive and usable for all users, including those with disabilities.

6.3. BACK-END

6.3.1. PHP:

PHP, which stands for Hypertext Preprocessor, is a widely-used server-side scripting language primarily designed for web development. It's often embedded into HTML to add dynamic features to web pages or interact with databases. PHP code is executed on the server, generating HTML content that is then sent to the client's web browser. It's known for its simplicity, versatility, and compatibility with various databases and web servers. PHP is commonly used for tasks like handling form data, generating dynamic page content, managing user sessions, and interacting with databases, making it a fundamental component of many websites and web applications.

- 1. Processing Form Data:** PHP handles user submissions like orders and payment info, validating and securely storing the data.
- 2. Generating Dynamic Content:** PHP fetches and displays real-time content like menus and promotions from databases, ensuring users see updated information.
- 3. Managing User Sessions:** PHP manages user logins and personalized features, maintaining session data for seamless browsing experiences.
- 4. Interacting with Databases:** PHP communicates with databases to store and retrieve essential data, ensuring consistency and integrity.
- 5. Handling Business Logic:** PHP executes critical business operations such as calculating totals, validating inputs, and managing discounts.

6.4. DATABASE MANAGEMENT

6.4.1. MySQL:

MySQL is a popular open-source relational database management system (RDBMS) that is widely used in web development. It provides a powerful and scalable platform for storing, organizing, and retrieving data in a structured manner. MySQL uses SQL (Structured Query Language) as its primary interface for managing databases, allowing users to perform operations such as querying, updating, and deleting data.

- 1. Storing User Data:** MySQL is used to store user information such as login credentials, profile details, and order history. This data allows users to log in, manage their accounts, and access personalized features.
- 2. Recording Orders:** MySQL records details of user orders, including selected items, quantities, special instructions, and order statuses. This data facilitates order processing, tracking, and fulfillment, ensuring efficient handling of customer requests.
- 3. Managing Restaurant Information:** MySQL stores information about partner restaurants, including names, addresses, contact details, operating hours, and delivery areas. This data helps users find nearby restaurants, view their details, and place orders for delivery or pickup.
- 4. Tracking Delivery Information:** MySQL stores delivery-related information such as customer addresses, delivery statuses, delivery personnel assignments, and delivery times. This data enables real-time tracking of delivery orders and ensures timely and accurate delivery to customers.

6.5. PURPOSE

The purpose of an online food ordering system projects are multifaceted, aimed at enhancing the overall dining experience for both users and restaurant owners. The project aims to provide users with a convenient and accessible platform to browse through a diverse range of restaurants and cuisines, place orders, and arrange delivery or pickup options from the comfort of their homes or workplaces. This convenience factor is essential in today's fast-paced world, catering to busy lifestyles and offering a seamless solution for satisfying culinary cravings. Secondly, the project seeks to streamline the ordering process, minimizing wait times, eliminating the need for phone calls, and facilitating quick and hassle-free transactions, thereby enhancing operational efficiency for restaurants and reducing errors in order processing.

- 1. Convenience:** The online food ordering system project aims to offer users a convenient way to browse menus, place orders, and arrange delivery or pickup from a wide range of restaurants and cuisines. By providing an easy-to-use platform accessible from any device with internet connectivity, the project ensures that users can satisfy their culinary cravings without the hassle of visiting restaurants or making phone calls.
- 2. Efficiency:** One of the primary purposes of the project is to streamline the ordering process, reducing wait times and errors in order processing. By digitizing the ordering process, the project minimizes the manual effort required by both users and restaurant staff, leading to faster transactions, improved order accuracy, and enhanced operational efficiency for restaurants.

- 3. Transparency:** The project seeks to foster transparency and trust by providing users with comprehensive information about restaurants, menus, prices, delivery options, and order statuses. By offering transparency at every stage of the ordering process, the project ensures that users can make informed decisions and trust the platform for their dining needs.
- 4. Accessibility:** Another key purpose of the project is to make food ordering accessible to a wide range of users, including those with disabilities or limited mobility. By providing accessible interfaces and options for customization, the project ensures that all users can easily navigate the platform and place orders without barriers.
- 5. Innovation:** The project aims to innovate the dining experience by leveraging technology such as artificial intelligence (AI), machine learning, and augmented reality (AR). By introducing features such as personalized recommendations, predictive analytics for inventory management, and immersive dining experiences, the project seeks to stay ahead of trends and provide users with unique and engaging experiences.
- 6. Business Growth:** Ultimately, the project aims to drive business growth for restaurants by expanding their reach, increasing order volumes, and improving customer satisfaction. By providing restaurants with a reliable platform to showcase their offerings and efficiently manage orders, the project contributes to the success and sustainability of the food service industry.

6.6. SYSTEM ARCHITECTURE

In the context of an online food ordering system, the system architecture refers to the high-level structure and components that enable the functionality and operation of the platform. The system architecture typically comprises several layers, each serving a specific purpose and interacting with other layers to fulfill user requests and process data effectively.

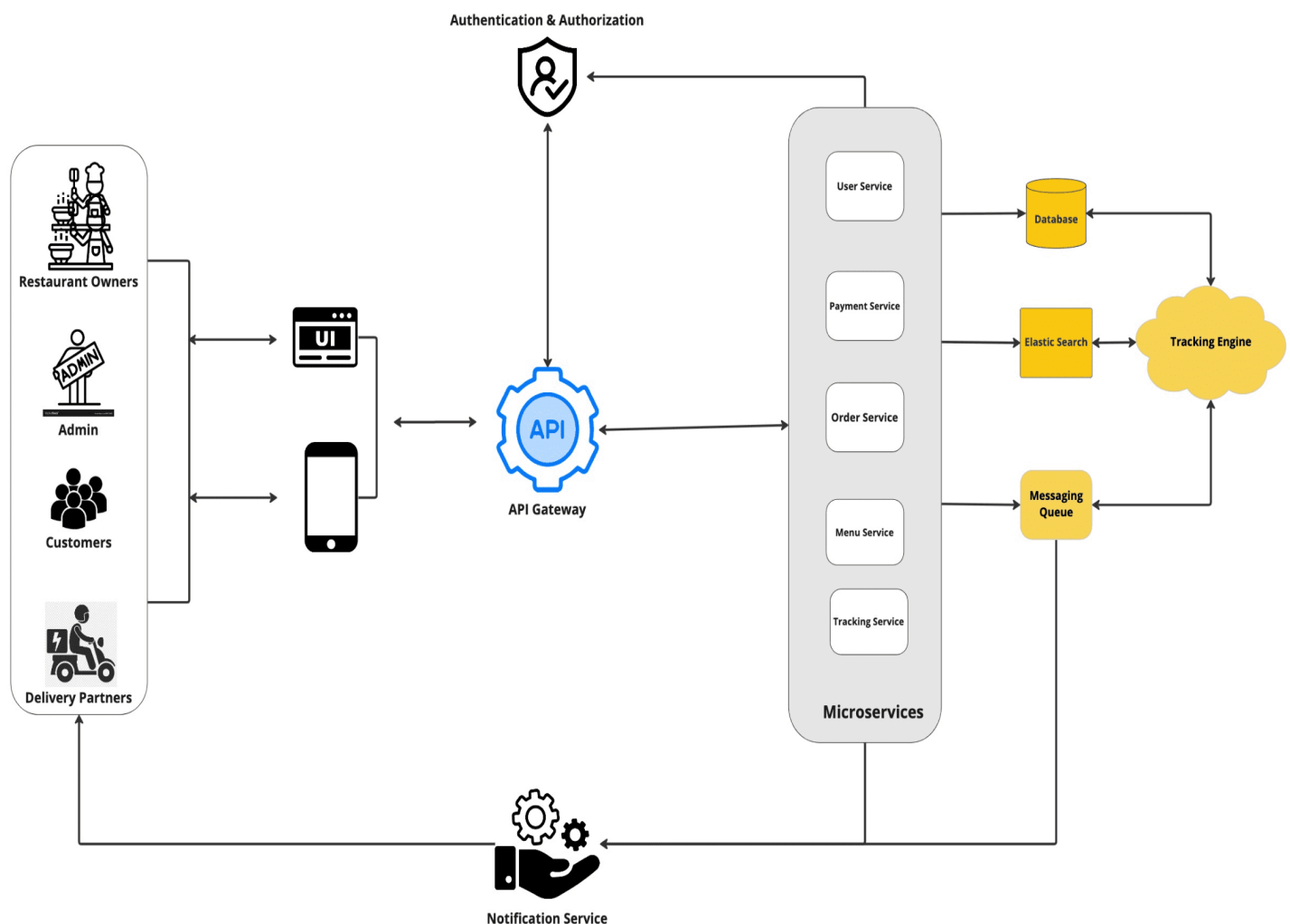


Figure 1: System Architecture of Online Food Ordering System

- 1. Presentation Layer:** This layer represents the user interface of the online food ordering system, including web or mobile applications used by customers to browse restaurants, view menus, place orders, and track deliveries. It focuses on providing an intuitive and responsive interface that enhances user experience and facilitates interaction with the system.
- 2. Application Layer:** The application layer contains the business logic and functionality of the online food ordering system. It handles user requests, processes orders, manages restaurant listings and menus, handles payment transactions, and coordinates interactions between different system components. This layer implements the core features of the platform and ensures smooth operation of the ordering process.
- 3. Data Layer:** The data layer is responsible for managing and storing data related to users, restaurants, menus, orders, payments, and other relevant information. It typically includes a relational database management system (RDBMS) such as MySQL, PostgreSQL, or MongoDB, where data is organized, stored, and retrieved efficiently. The data layer ensures data integrity, security, and scalability to support the growing needs of the platform.
- 4. Integration Layer:** The integration layer facilitates communication and integration with external systems and services, such as payment gateways, delivery logistics providers, and third-party APIs. It enables seamless interactions between the online food ordering system and external entities, ensuring smooth operation and extending the platform's functionality through integrations with external services.

6.7. METHODOLOGY

1. Problem Understanding:

Begin by thoroughly understanding the requirements and challenges of online food ordering, including user needs, business objectives, market trends, and regulatory considerations. Conduct stakeholder interviews, surveys, and market research to gain insights into the problem domain.

2. Research and Familiarization:

Conduct extensive research to explore existing online food ordering systems, industry best practices, emerging technologies, and potential competitors. Familiarize yourself with relevant programming languages, frameworks, and tools commonly used in this domain.

3. Environment Setup:

Set up the development environment by configuring servers, databases, development tools, and version control systems. Create a robust infrastructure for hosting and deploying the online food ordering system, ensuring scalability, security, and reliability.

4. Code Implementation:

Develop the online food ordering system according to the identified requirements and design specifications. Use appropriate programming languages, frameworks, and architectural patterns to implement features such as user authentication, menu management, order processing, and payment integration.

5. Testing and Validation:

Conduct thorough testing to validate the functionality, usability, security, and performance of the system. Perform unit testing, integration testing, and system testing to identify and fix defects, ensuring that the system meets quality standards and user expectations.

6. Performance Optimization:

Optimize the performance of the online food ordering system by analyzing bottlenecks, optimizing database queries, caching frequently accessed data, and implementing techniques such as lazy loading and code minification. Monitor system performance and scalability to ensure optimal user experience under varying loads.

7. Documentation and Reporting:

Document the design, architecture, implementation details, and testing procedures of the online food ordering system. Prepare comprehensive user manuals, technical documentation, and reports to facilitate system maintenance, troubleshooting, and knowledge transfer.

8. Future Enhancements:

Continuously monitor user feedback, market trends, and technological advancements to identify opportunities for future enhancements and optimizations. Prioritize and plan future development initiatives to add new features, improve user experience, and stay ahead of competition in the dynamic online food ordering market.

6.8. DEVELOPMENT TOOLS

1. Database Environment:

- A robust database environment is crucial for storing and managing data related to users, restaurants, menus, orders, payments, and more in the online food ordering system.
- Tools like MySQL, PostgreSQL, or MongoDB provide features for creating, querying, updating, and maintaining databases. They ensure data integrity, security, and scalability, allowing the system to handle large volumes of transactions.

2. Web Technology:

- Web technologies are the foundation of the online food ordering system, powering the frontend user interface and backend server-side logic.
- HTML, CSS, and JavaScript are fundamental for building responsive and interactive user interfaces. Frameworks and libraries like React, Angular, or Vue.js streamline frontend development, while server-side technologies like Node.js, Django, or Ruby on Rails facilitate backend development.

3. System Platform:

- Choosing the right system platform is essential for deploying and running the online food ordering system reliably and securely.
- Cloud platforms such as AWS, Azure, or Google Cloud offer scalable infrastructure, managed services, and security features for hosting web

applications. They provide resources like virtual servers, databases, storage, and networking, enabling seamless deployment and management of the system.

4. Project Management Tool:

- Project management tools help organize tasks, track progress, collaborate with team members, and ensure timely delivery of project milestones.
- Tools like Jira, Trello, or Asana facilitate task management, issue tracking, sprint planning, and communication among team members. They provide features such as Kanban boards, Gantt charts, and workflow automation to streamline project workflows and enhance productivity.

5. Visual Paradigm Community Editor:

- Visual modeling tools are valuable for designing system architectures, data models, and software diagrams to visualize and communicate project requirements and designs.
- Visual Paradigm Community Edition offers a range of diagramming tools for creating UML diagrams, ER diagrams, flowcharts, and more. It helps developers and stakeholders understand system components, relationships, and behaviors, aiding in design decisions and implementation planning.

6. Test Plan:

- Testing is critical for ensuring the quality, reliability, and functionality of the online food ordering system before deployment. After the system has been developed, it will move to the system testing phase. In system testing phase, the developed system is required to install on appropriate devices for testing purpose. After the system install has been completed, the system testing task will be performed by different roles of user such as manager role and staff role.
 - i. **Unit Testing:** Involves testing individual components or modules of the system in isolation to verify their correctness and functionality. Tools like Jest, Mocha, or PHPUnit automate unit tests, allowing developers to catch and fix bugs early in the development cycle.
 - ii. **Functional Testing:** Focuses on validating the system's functionality against specified requirements and use cases. Tools like Selenium, Cypress, or Puppeteer automate functional tests by simulating user interactions and verifying expected outcomes, ensuring the system behaves as intended across different scenarios.

6.9. TESTING

1. Unit Testing:

- **Purpose:** Unit testing verifies the correctness of individual components or modules of the system in isolation.
- **Approach:** Developers write test cases to validate the behavior of functions, classes, or methods, covering different scenarios and edge cases.
- **Tools:** Common unit testing frameworks include Jest, Mocha, PHPUnit, and Jasmine.
- **Example:** Testing functions that handle user authentication, menu retrieval, order processing, and payment validation to ensure they work as expected.

2. Integration Testing:

- **Purpose:** Integration testing verifies the interactions between different components or modules of the system.
- **Approach:** Developers test how components interact with each other, ensuring seamless communication and data flow.
- **Tools:** Integration testing frameworks such as Supertest, Flask-Testing, and Django's TestCase module.
- **Example:** Testing the integration between the frontend user interface and backend API endpoints to ensure proper data transmission and error handling.

3. System Testing:

- **Purpose:** System testing validates the entire online food ordering system as a whole.
- **Approach:** Testers simulate real-world user interactions and scenarios to verify the system's functionality, performance, and usability.
- **Tools:** Automated testing tools like Selenium, Cypress, and Puppeteer for end-to-end testing.
- **Example:** Testing the complete ordering process, including user registration, menu browsing, order placement, payment processing, and order tracking, to ensure all features work seamlessly together.

4. Acceptance Testing:

- **Purpose:** Acceptance testing verifies that the system meets the specified requirements and is ready for deployment.
- **Approach:** Testers execute predefined test cases based on user stories, use cases, and acceptance criteria to validate the system's functionality from an end-user perspective.
- **Tools:** Manual testing by stakeholders, automated acceptance testing tools like Cucumber or Behave.
- **Example:** Testing the online food ordering system with real users or stakeholders to ensure it meets their expectations in terms of usability, performance, and business requirements.

6.10. SYSTEM LIMITATIONS

One significant limitation of online food ordering systems is their susceptibility to technical issues and disruptions. These systems heavily rely on technology, making them vulnerable to website crashes, slow loading times, and payment processing errors, which can impede the user experience and lead to frustration among customers. Additionally, the dependency on internet connectivity poses challenges in areas with poor network coverage or during network outages, hindering users' ability to access the platform and place orders seamlessly. These technical limitations underscore the importance of robust infrastructure, continuous monitoring, and proactive measures to ensure the reliability, stability, and performance of online food ordering systems.

6.11. SYSTEM STRENGTHS

An online food ordering system offers numerous strengths that revolutionize the dining experience for users and businesses alike. Foremost among these is unparalleled convenience, as users can effortlessly browse through a vast selection of restaurants and cuisines, place orders from the comfort of their homes or workplaces, and enjoy swift delivery or pickup options. The accessibility of the system ensures that users have access to diverse culinary options, catering to varying tastes and dietary preferences. Furthermore, the system streamlines the ordering process, minimizing wait times, eliminating the need for phone calls, and facilitating quick and hassle-free transactions, particularly during peak hours. These strengths collectively enhance user satisfaction, increase operational efficiency for restaurants, and contribute to the overall growth and success of the online food ordering ecosystem.

7. DESIGN

7.1. Introduction:

The design phase of the online food ordering system involves conceptualizing the architecture, user interface, and functionality of the system. It lays the foundation for the development process by outlining the system's components, interactions, and workflows.

7.2. UML Diagrams:

UML (Unified Modeling Language) diagrams are used to visually represent the system's structure and behavior. This includes diagrams such as:

- **Use Case Diagrams:** Illustrate the system's functionality from the perspective of different actors (e.g., users, administrators).
- **Class Diagrams:** Depict the static structure of the system, including classes, attributes, and relationships.
- **Sequence Diagrams:** Show the interactions between system components or actors over time, depicting the flow of messages or actions.
- **Activity Diagrams:** Represent the workflow or process flow within the system, detailing the sequence of activities and decision points.

7.3. Use Case:

Use cases describe the interactions between users (actors) and the system to achieve specific goals or tasks. In the context of an online food ordering system, typical use cases include:

- **Registering/Login:** Users create accounts or log in to access the system.

- **Browsing Menu:** Users explore restaurant menus and view dish details.
- **Placing Order:** Users select items, customize orders, and proceed to checkout.
- **Managing Profile:** Users update personal information, preferences, or payment methods.
- **Admin Management:** Administrators manage restaurants, menus, orders, and user accounts.
- **Order Tracking:** Users track the status of their orders in real-time.

There are various kinds of methods in software design:

They are as follows:

- Use case diagram
- Sequence diagram
- Collaboration diagram
- Activity diagram
- State chat diagram

7.4. USE CASE DIAGRAMS

Use case diagrams model behavior within a system and helps the developers understand of what the user requires. The stickman represents what's called an actor.

Use case diagram can be useful for getting an overall view of the system and clarifying who can do and more importantly what they can't do.

Use case diagrams consists of use cases and actors and shows the interaction between the use case and actors.

- The purpose is to show the interactions between the use case and actor.
- To represent the system requirements from user's perspective.
- An actor could be the end-user of the system or an external system.

Use-case diagrams are helpful in the following situations:

- Before starting a project, you can create use-case diagrams to model a business so that all participants in the project share an understanding of the workers, customers, and activities of the business.
- While gathering requirements, you can create use-case diagrams to capture the system requirements and to present to others what the system should do.
- During the analysis and design phases, you can use the use cases and actors from your use-case diagrams to identify the classes that the system requires.
- During the testing phase, you can use use-case diagrams to identify tests for the system.

7.4.1. Admin Use Case Diagram:

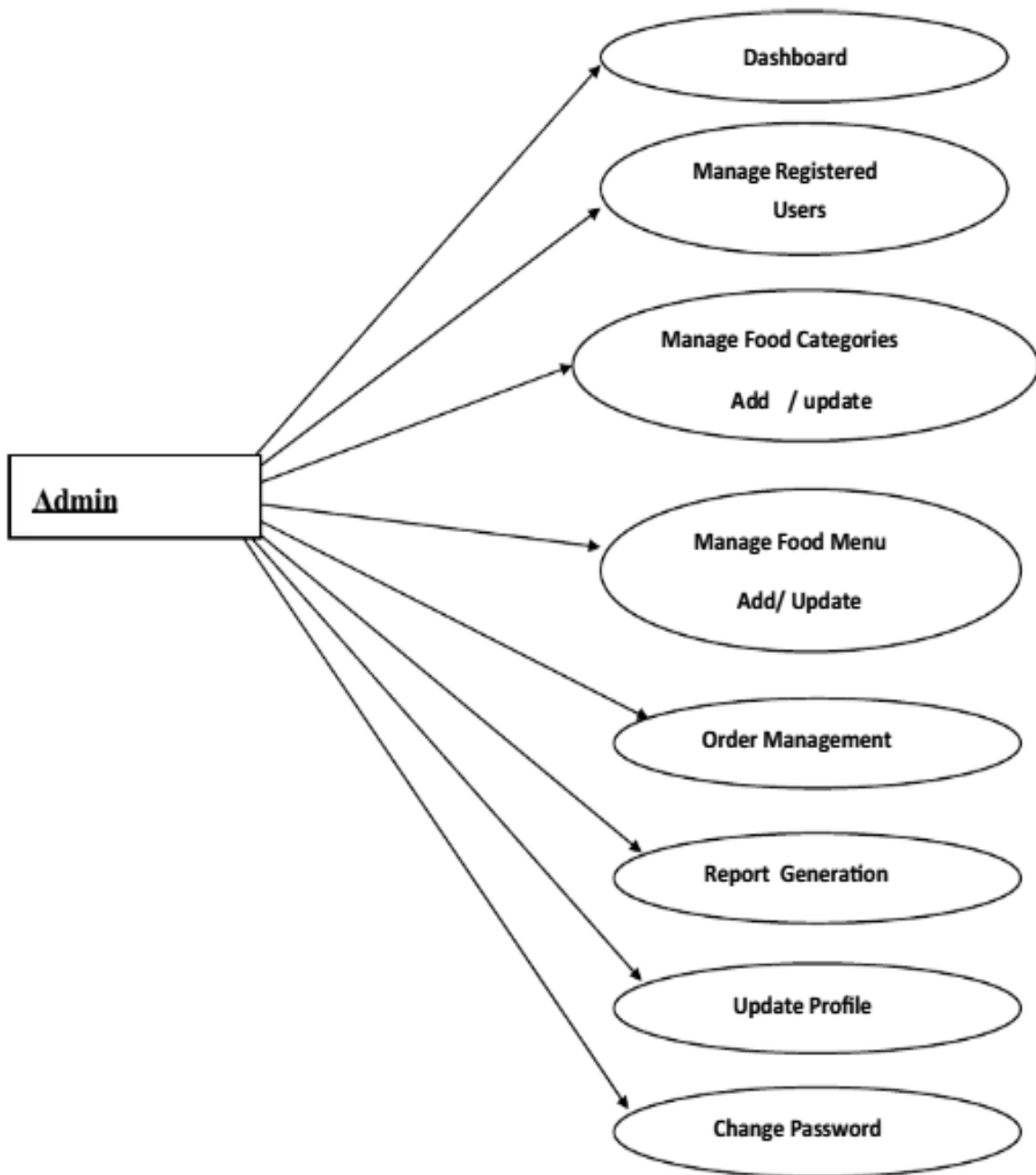


Figure 2: Diagram for Admin Use Case

7.4.2. User Use Case Diagram:

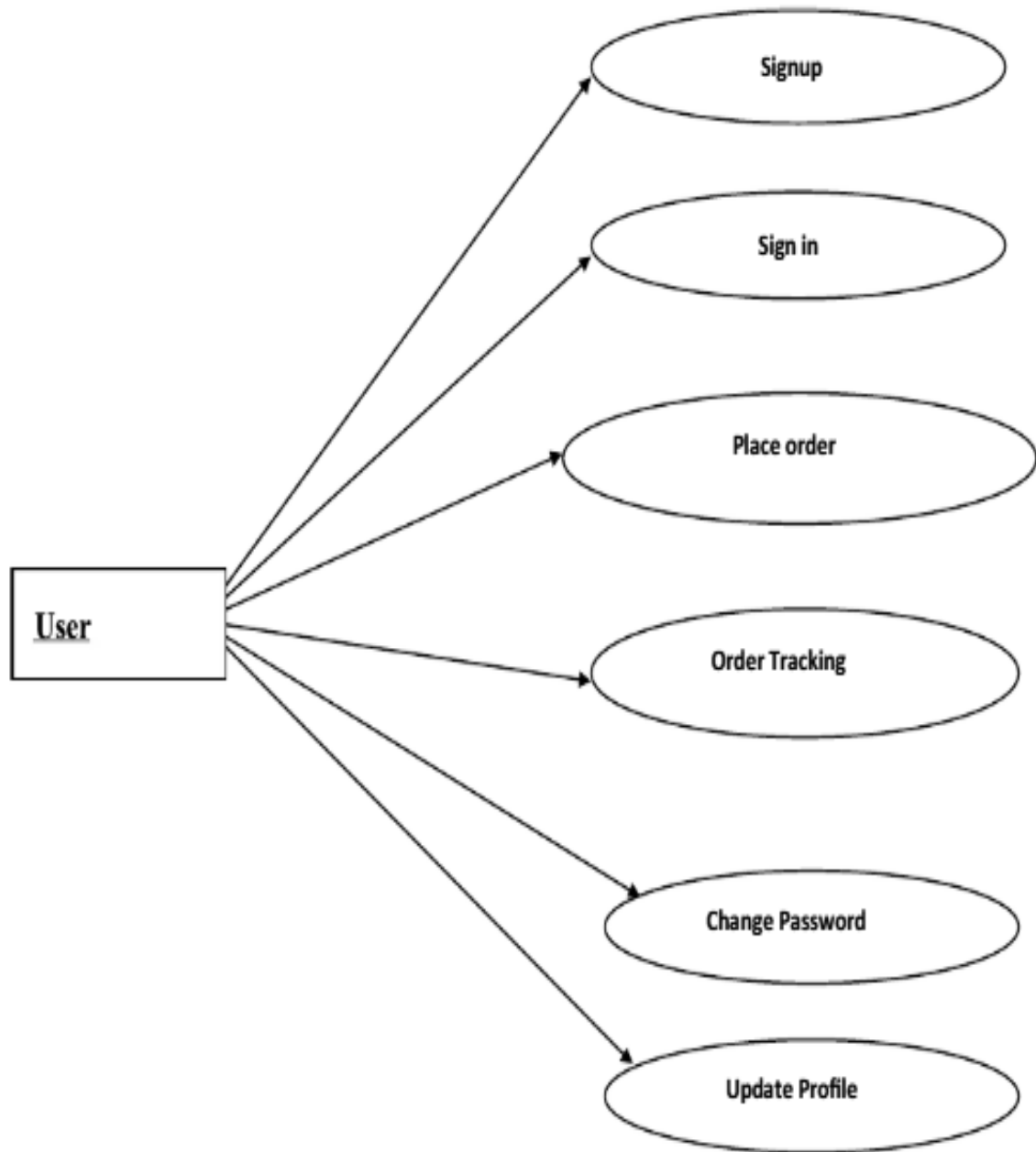


Figure 3: Diagram for User Use Case

7.5. DATA FLOW DIAGRAM

A Data Flow Diagram (DFD) is a graphical representation of the flow of data within a system or process. In the context of an online food ordering system, a Data Flow Diagram (DFD) is a visual representation that illustrates the flow of data within the system. It depicts how data moves between various processes, entities, and data stores, providing a high-level overview of the system's functionality and data interactions.

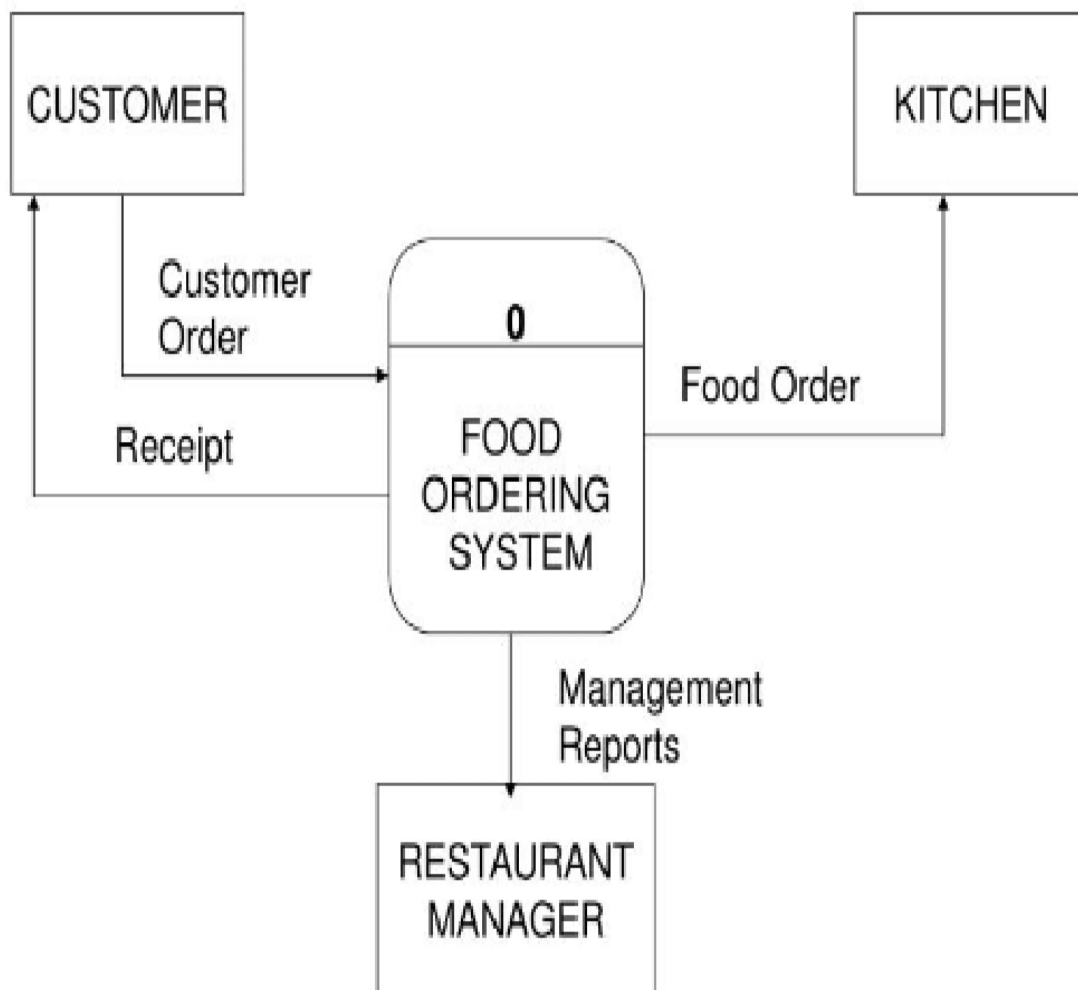


Figure 4: DFD Diagram for Online Food Ordering System

1. External Entities:

- **Users:** Represented as external entities, users interact with the system by submitting orders, browsing menus, and providing feedback.
- **Restaurants:** External entities that provide menu listings, receive orders, and update order statuses.

2. Processes:

- **Order Placement:** This process involves users browsing menus, selecting items, customizing orders, and submitting them for processing.
- **Order Processing:** Once an order is placed, this process verifies payment, updates inventory, notifies the restaurant, and generates order confirmations for users.
- **Menu Management:** This process involves updating menu listings, adding new dishes, adjusting prices, and managing availability based on inventory levels.
- **Feedback Handling:** This process manages user feedback and reviews, collecting and storing feedback data for analysis and improvement.

3. Data Stores:

- **User Database:** Stores user profiles, preferences, order history, and authentication credentials.

- **Menu Database:** Stores information about restaurant menus, including dish names, descriptions, prices, and availability.
- **Order Database:** Stores order details, including items ordered, quantities, prices, delivery addresses, and order statuses.
- **Feedback Database:** Stores user feedback, ratings, reviews, and comments associated with orders and restaurant experiences.

4. Data Flows:

- **Order Data Flow:** Represents the flow of order information from users to the system, including order details, preferences, and delivery instructions.
- **Menu Data Flow:** Represents the flow of menu information from restaurants to the system, including menu listings, dish details, and pricing.
- **Feedback Data Flow:** Represents the flow of feedback information from users to the system, including ratings, reviews, and comments.
- **Notification Data Flow:** Represents the flow of notification information from the system to users and restaurants, including order confirmations, updates, and alerts.

7.6. E-R DIAGRAM

In the context of an online food ordering system, an Entity-Relationship (E-R) diagram is a visual representation that depicts the entities, attributes, and relationships within the system's database. It illustrates the structure of the database and how different entities are related to each other.

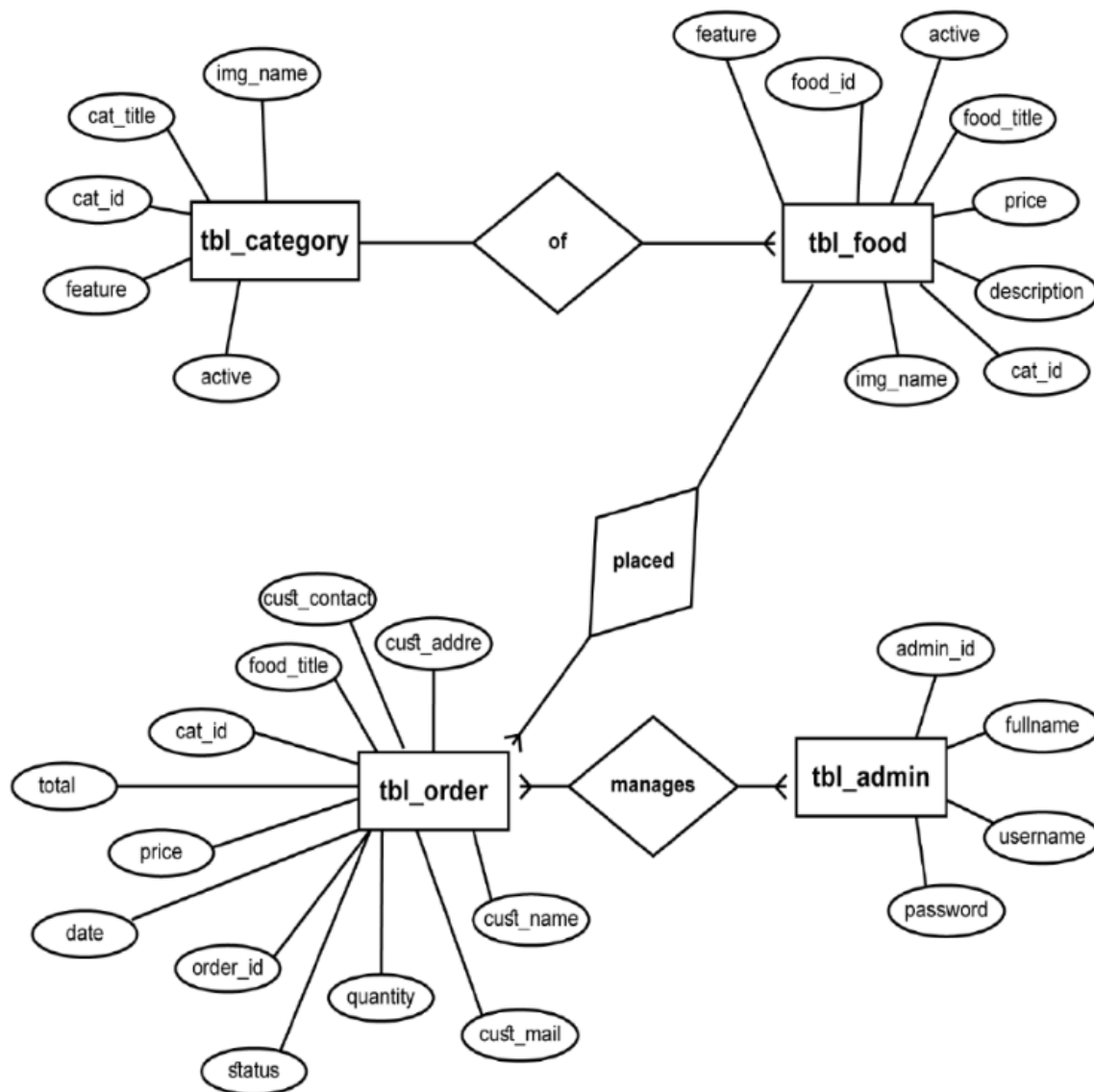


Figure 5: E-R Diagram for Online Food Ordering System

1. Entities:

- **User:** Represents individuals who interact with the system by browsing menus, placing orders, and providing feedback. Attributes may include user ID, name, contact information, and preferences.
- **Restaurant:** Represents establishments that offer food items for ordering. Attributes may include restaurant ID, name, location, contact information, and operating hours.
- **Menu Item:** Represents individual dishes or items available for ordering from restaurants. Attributes may include item ID, name, description, price, and availability status.
- **Order:** Represents a transaction initiated by a user to purchase food items from a restaurant. Attributes may include order ID, user ID, restaurant ID, order status, total amount, and delivery details.
- **Feedback:** Represents user feedback and reviews submitted after completing an order. Attributes may include feedback ID, user ID, restaurant ID, rating, review text, and timestamp.

2. Relationships:

- **User-Order:** Represents the relationship between users and orders, indicating that a user can place multiple orders, but each order is associated with a single user.
- **Restaurant-Menu Item:** Represents the relationship between restaurants and menu items, indicating that a restaurant can offer

multiple menu items, but each menu item is associated with a single restaurant.

- **Order-Menu Item:** Represents the relationship between orders and menu items, indicating that an order can include multiple menu items, and each menu item can be part of multiple orders.
- **User-Feedback:** Represents the relationship between users and feedback, indicating that a user can submit multiple feedback entries, but each feedback entry is associated with a single user.

3. Cardinality and Constraints:

- Cardinality notations such as "one-to-many" and "many-to-many" depict the relationship between entities, indicating the number of instances of one entity that can be associated with another entity.
- Constraints such as "mandatory" or "optional" specify whether a relationship is required or optional, and whether it can exist independently of other entities.

7.7. FLOWCHART

The online food ordering system begins with user login or signup, followed by browsing the menu and selecting items, which are then added to the cart. After reviewing the cart, the user proceeds to checkout, enters delivery details, and chooses a payment method. Upon payment processing, an order confirmation is generated, and the restaurant is notified. The restaurant prepares the order, which is then picked up by a delivery person for delivery to the user. Once the user receives the order, the process concludes.

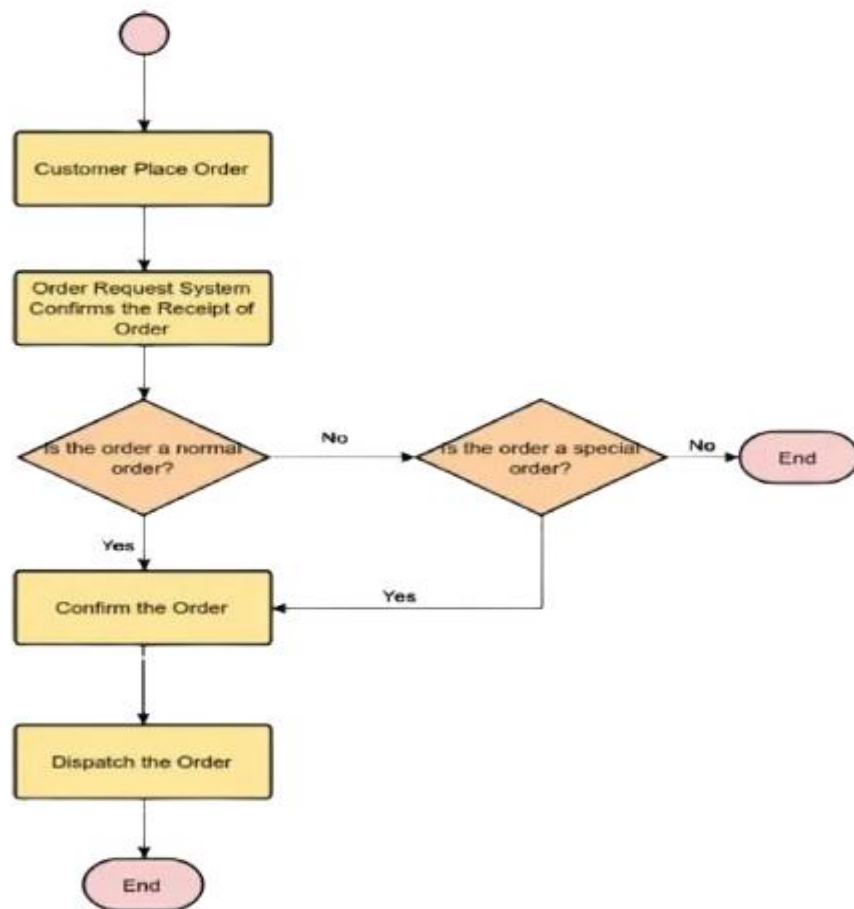


Figure 6: Flowchart Diagram for Food Ordering System

7.8. DATABASE DESIGN

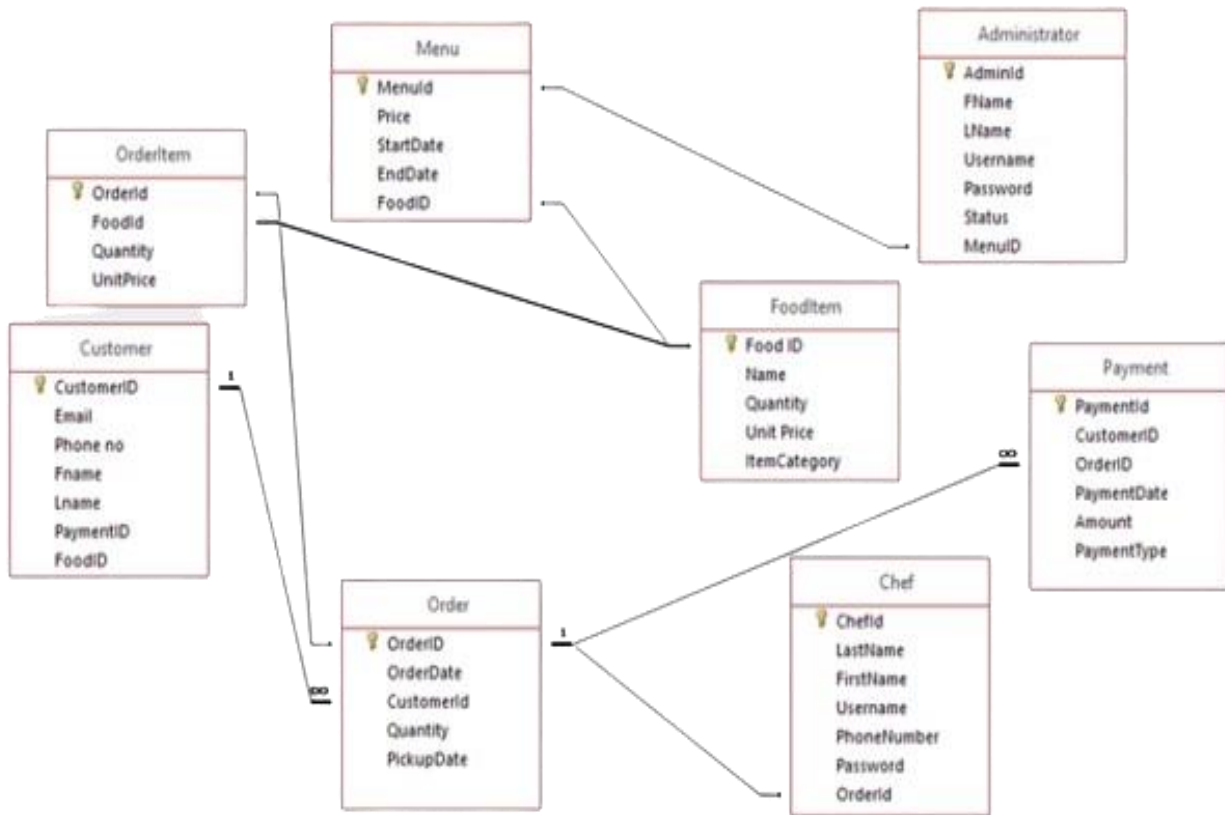


Figure 7: Database Design for Food Ordering System

7.8.1. Database Tables:

Table	Action	Rows	Type	Collation	Size	Overhead
☐ admin	Browse Structure Search Insert Empty Drop	1	InnoDB	utf8mb4_general_ci	16.0 KiB	-
☐ cart	Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_general_ci	16.0 KiB	-
☐ messages	Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_general_ci	16.0 KiB	-
☐ orders	Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_general_ci	16.0 KiB	-
☐ products	Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_general_ci	16.0 KiB	-
☐ users	Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_general_ci	16.0 KiB	-
6 tables	Sum	1	InnoDB	utf8mb4_general_ci	96.0 KiB	0 B

Figure 8: Database Table Schema

1. Admin Table
2. Cart Table
3. Messages Table
4. Orders Table
5. Products Table
6. Users Table

In an online food ordering system, database tables store various kinds of data needed to support the functionalities of the platform. These tables are essential for organizing data efficiently and ensuring that the system runs smoothly. Below is a detailed explanation of specific tables and their role within an online food ordering system:

1. Admin Table:

The Admin Table manages information about the administrators who have the authority to manage the system. Admins can perform tasks such as managing users, products, and orders.

Columns:

- **admin_id** (Primary Key): Unique identifier for each admin.
- **username**: Admin's username.
- **password_hash**: Hashed password for security.
- **email**: Admin's email address.
- **created_at**: Timestamp of when the admin account was created.

- **updated_at**: Timestamp of the last update to the admin account.
- **role**: Role of the admin, which determines the level of access (e.g., super admin, support admin).

2. Cart Table:

The Cart Table holds information about the items users have added to their shopping carts but have not yet purchased. It helps in managing the user's current order before checkout.

Columns:

- **cart_id** (Primary Key): Unique identifier for each cart entry.
- **user_id** (Foreign Key): Identifier of the user who owns the cart.
- **product_id** (Foreign Key): Identifier of the product added to the cart.
- **quantity**: Number of units of the product in the cart.
- **created_at**: Timestamp of when the item was added to the cart.
- **updated_at**: Timestamp of the last update to the cart entry.

3. Messages Table:

The Messages Table stores messages exchanged between users and the system or between users and administrators. This can include customer support inquiries, order updates, and other communications.

Columns:

- **message_id** (Primary Key): Unique identifier for each message.
- **sender_id** (Foreign Key): Identifier of the sender (could be a user or admin).

- **receiver_id** (Foreign Key): Identifier of the receiver (could be a user or admin).
- **message_text**: Content of the message.
- **created_at**: Timestamp of when the message was sent.

4. Orders Table:

The Orders Table manages information about orders placed by users. It tracks the details of each order from placement to delivery.

Columns:

- **order_id** (Primary Key): Unique identifier for each order.
- **user_id** (Foreign Key): Identifier of the user who placed the order.
- **total_amount**: Total amount of the order.
- **order_status**: Current status of the order (e.g., pending, confirmed, prepared, delivered).
- **delivery_address**: Address where the order should be delivered.
- **created_at**: Timestamp of when the order was placed.
- **updated_at**: Timestamp of the last update to the order.
- **payment_method**: Method used for payment (e.g., credit card, cash on delivery).

5. Products Table:

The Products Table contains information about the food items available for order. This includes details such as name, description, price, and availability.

Columns:

- **product_id** (Primary Key): Unique identifier for each product.
- **name**: Name of the product.
- **description**: Description of the product.
- **price**: Price of the product.
- **stock_quantity**: Number of units available in stock.
- **created_at**: Timestamp of when the product was added.
- **updated_at**: Timestamp of the last update to the product.

6. Users Table:

The Users Table manages information about users of the online food ordering system. It stores user details required for authentication and personalized services.

Columns:

- **user_id** (Primary Key): Unique identifier for each user.
- **username**: Username of the user.
- **password_hash**: Hashed password for security.
- **email**: User's email address.
- **first_name**: User's first name.
- **last_name**: User's last name.
- **phone_number**: User's phone number for contact purposes.
- **address**: User's primary address for delivery.

- **created_at**: Timestamp of when the user account was created.
- **updated_at**: Timestamp of the last update to the user account.

Relationships Between Tables:

1. **Users and Admin**: Admins manage the system, while users place orders and interact with the system. They might be managed as separate entities or admins can be a subset of users with special roles.
2. **Users and Cart**: Each user can have one or more carts with items they intend to purchase.
3. **Users and Orders**: Each user can place multiple orders.
4. **Users and Messages**: Users can send messages to the system (customer support) or to other users.
5. **Cart and Products**: Each cart entry corresponds to a specific product, with details about quantity.
6. **Orders and Products**: Orders usually reference products to indicate what was purchased, often managed through an intermediary table in many-to-many relationships (order details).

These tables form the backbone of an online food ordering system, ensuring that the application can efficiently manage users, products, orders, and interactions.

7.9. SNAPSHOTS OF THE PROJECT

7.9.1. HOME PAGE

Main landing page where users can access key features such as browsing restaurants, exploring menus, placing orders, viewing promotions, etc.

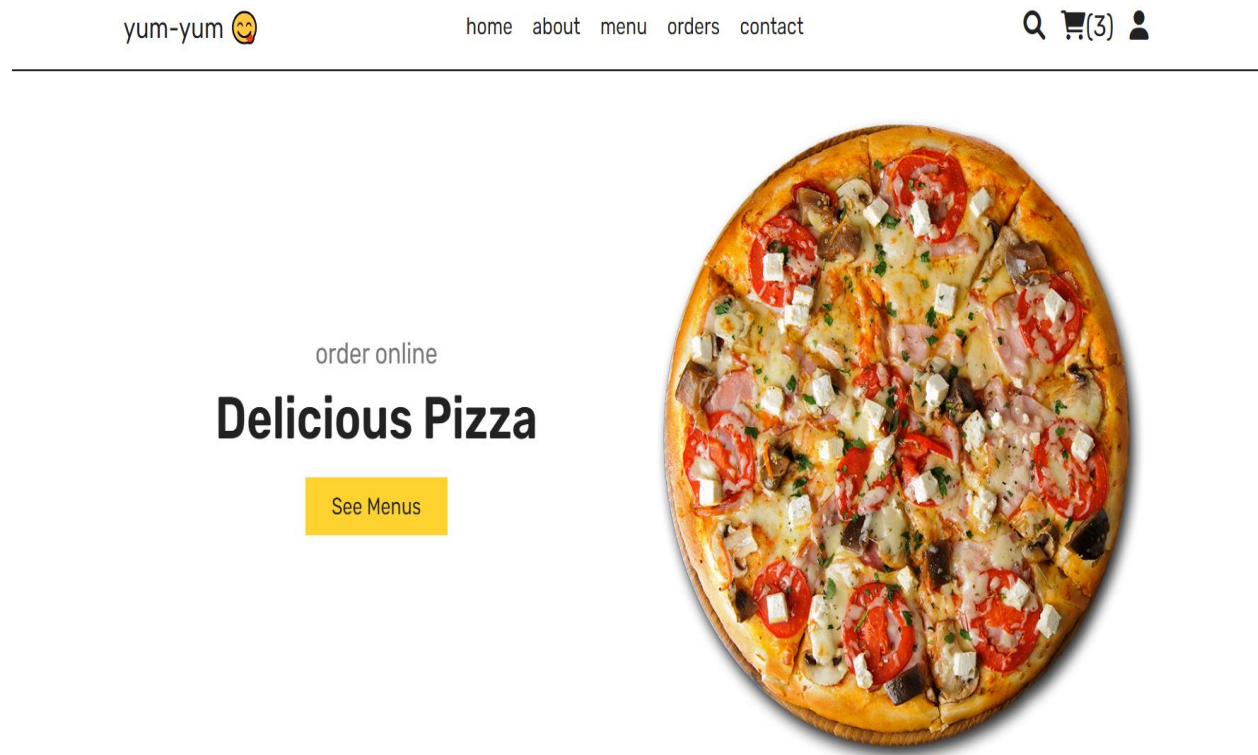


Figure 9: Website Homepage

7.9.2. CUSTOMER REGISTRATION PAGE

Creating new user accounts, enabling customers to sign up by providing their personal information such as name, email address, delivery address, and contact details.

yum-yum 😊
home about menu orders contact
🔍 🛒(0) 👤

REGISTER NOW

Register Now

already have an account? login now

Figure 10: User Registration Page

7.9.3. MENU PAGE

Displaying the available food items and offerings from participating restaurants, allowing customers to browse through various categories, view detailed descriptions, images, etc.

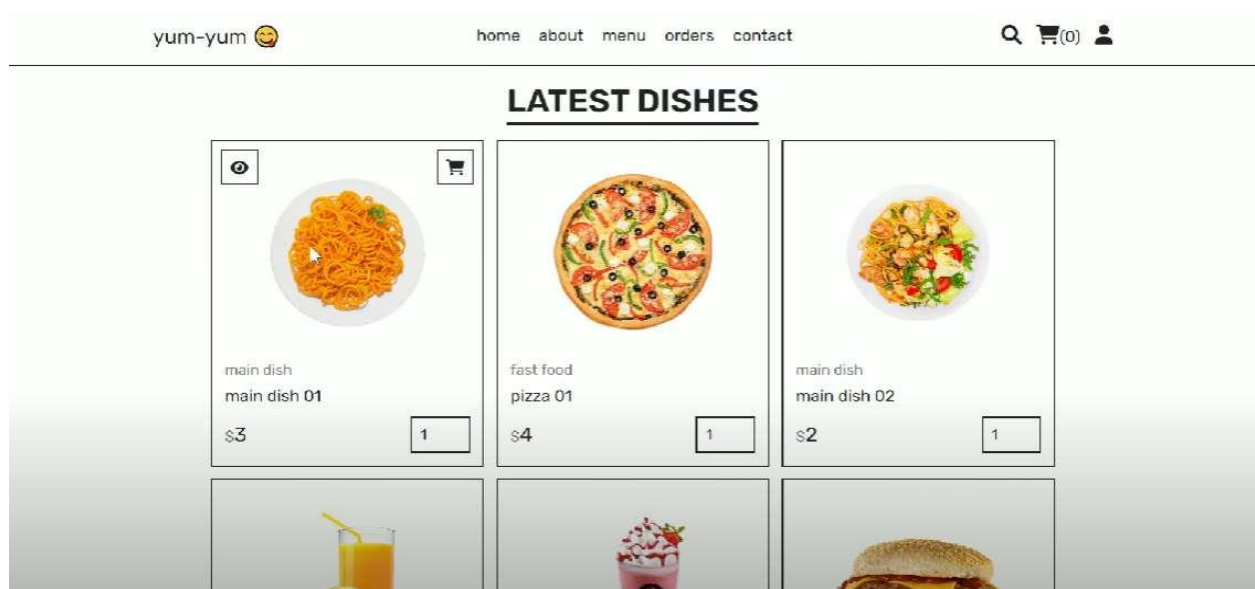


Figure 11: Food Menu Page

7.9.4. CONTACT US PAGE

A dedicated channel for customers to reach out to the platform's support team or customer service representatives.

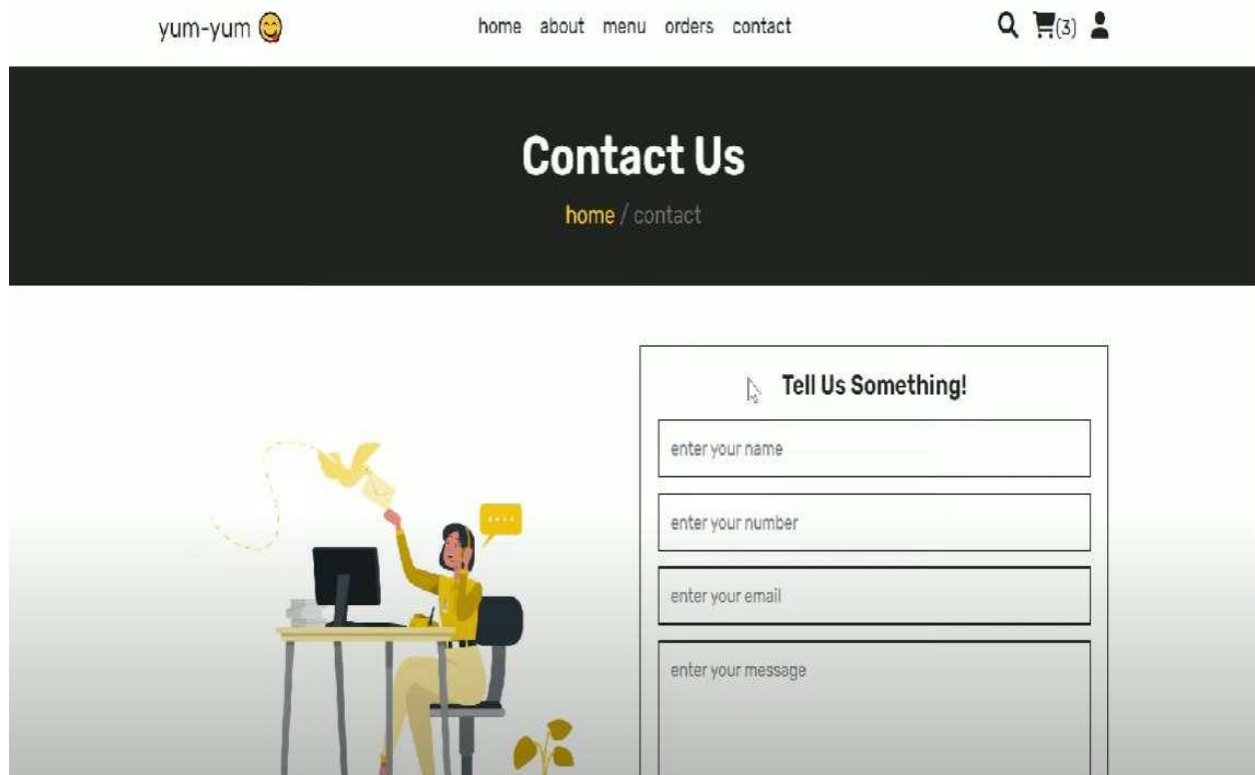


Figure 12: Contact Us Page

7.9.5. ADMIN PANEL

Admin Panel for managing various aspects of the platform, including menu updates, order management, user accounts, etc.

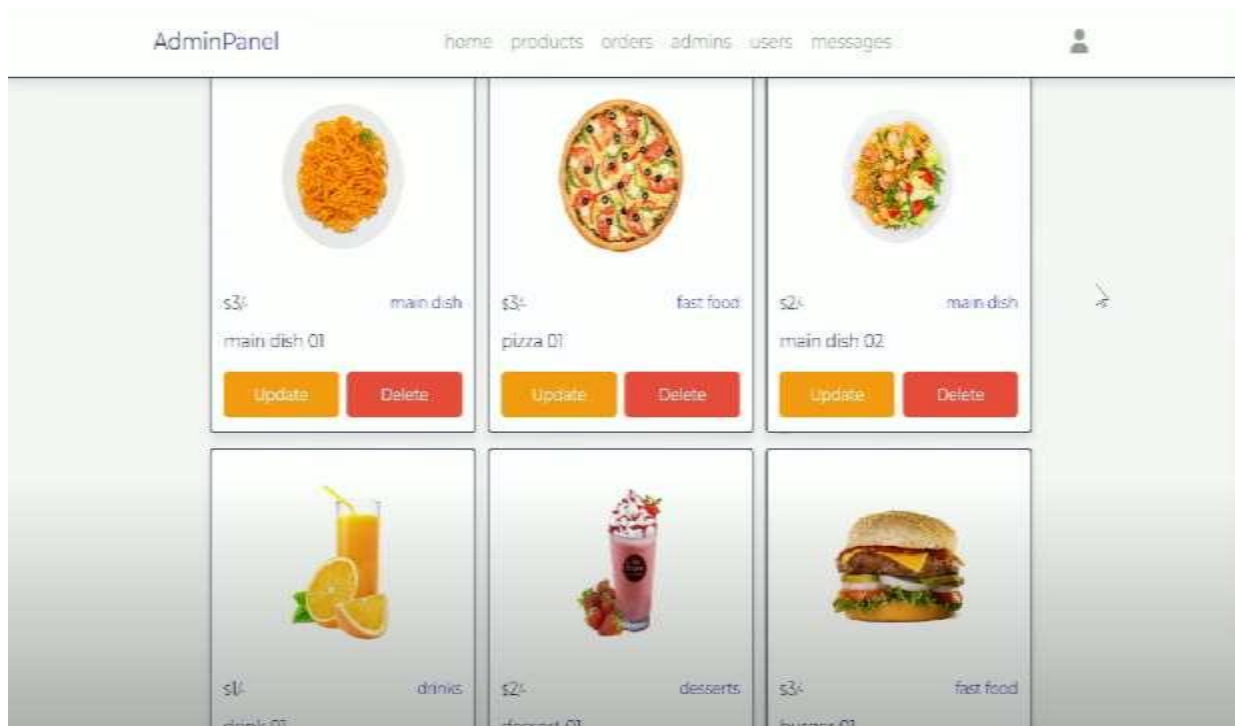
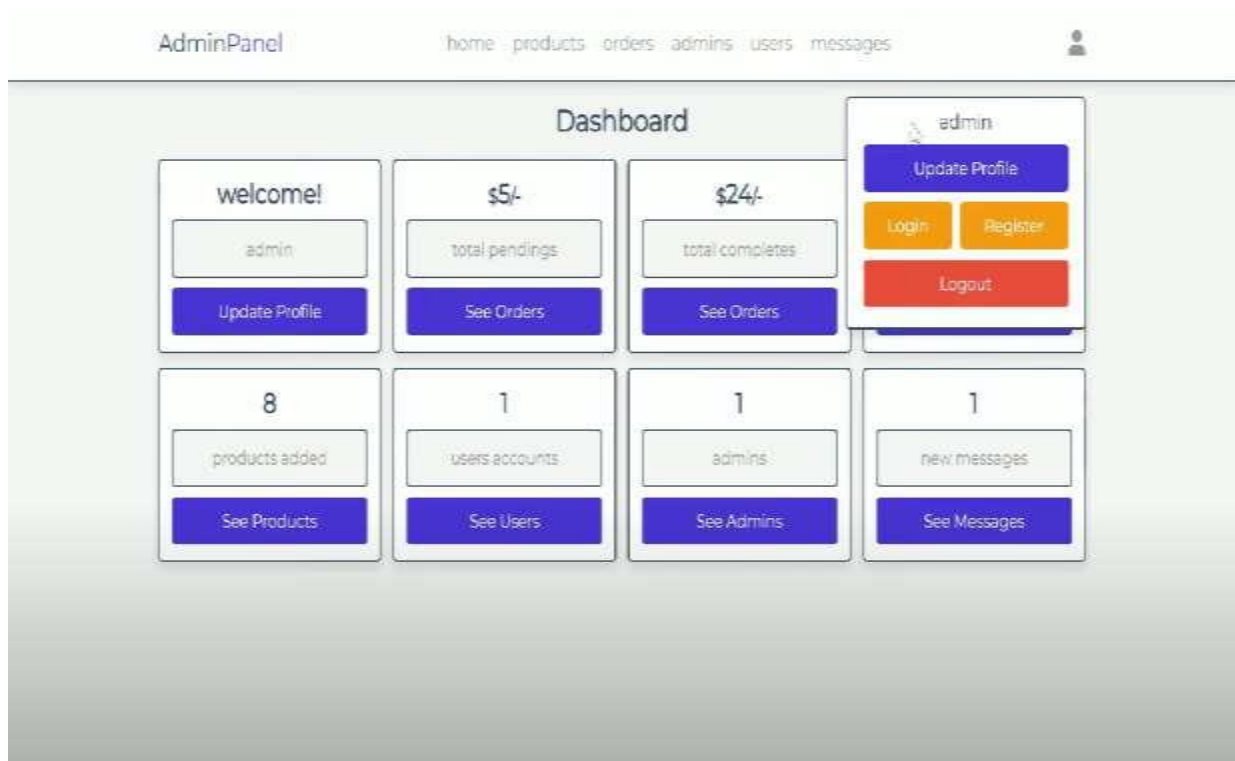


Figure 13: Admin Panel Page

8. CONCLUSION

After a decade, the advancement and innovation of technology helps people to manage their task easily and efficiently from just a click of a button. In many other industry areas have been used management system to assist their business grow long time ago, therefore it is a trend that cause F&B industry to make use of a management system for their business. At the end of this project the system can reduce and replace the human manpower task, reduce the time consume for each transaction and generate report for further management purpose by fully utilizing the system. Development of an online food ordering system represents a significant step forward in modernizing the dining experience for users and restaurant owners alike. Through the convenience of digital platforms, users can effortlessly explore a vast array of culinary options, place orders, and arrange delivery or pickup with ease. This system streamlines the ordering process, minimizing wait times, eliminating errors, and fostering transparency in transactions. Moreover, it provides restaurants with a platform to expand their reach, increase order volumes, and enhance customer satisfaction. The project's success hinges on robust infrastructure, intuitive user interfaces, seamless integration with external services, and a commitment to continuous improvement. As technology evolves and user preferences shift, the online food ordering system must adapt and innovate to meet evolving demands and remain competitive in the dynamic food service industry. Overall, the project signifies a transformative shift towards digitization and convenience in the realm of dining, promising a more efficient, accessible, and satisfying experience for all stakeholders involved.

9. FUTURE WORK

Future work for the online food ordering system involves continual innovation and adaptation to meet evolving user preferences and technological advancements. This includes enhancing personalization through advanced recommendation algorithms, integrating emerging technologies such as AI and AR to enhance user experience, expanding delivery options for greater convenience, ensuring accessibility for users with disabilities, optimizing sustainability practices, expanding into new markets for global growth.

Online food ordering systems is poised to be transformative, driven by advanced personalization through AI-driven recommendations, dynamic menu customization, and integration of dietary preferences, while enhancing user experience with voice-activated ordering, augmented reality menus, seamless payment options, and real-time order tracking. Operational efficiency will see significant improvements with automated order processing, sophisticated inventory management systems, and smart kitchen integration. Advanced analytics and insights, including predictive analytics, customer sentiment analysis, and business intelligence dashboards, will provide deeper understanding and optimization of operations. Sustainable practices such as eco-friendly packaging, carbon footprint tracking, and local sourcing initiatives will become integral, alongside the expansion of service capabilities like hyper-local delivery networks, multi-restaurant orders, subscription models, and virtual kitchens. Additionally, community engagement and social responsibility will be emphasized through partnerships with community-supported agriculture programs, charity donation features, and educational content on sustainable eating and nutrition, all contributing to a seamless, efficient, and socially responsible food ordering experience that supports both customers and food service providers.

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11. BIODATA OF EACH GROUP MEMBER

11.1. MR. SOHAIM ASLAM AZMI

Major:- Master of Computer Application

Email Id:- asohaim@student.iul.ac.in

Contact No.:- 7007252995

CAREER OBJECTIVE

To join a position of responsibility with a progressive organization that provides conducive learning and development.

EDUCATIONAL QUALIFICATION

Course	School/College	Board/University	Year of Passing	Marks/ Percentage
M.C.A	Integral University, Lucknow	Integral University	2024	Appearing
B.C.A	Integral University, Lucknow	Integral University	2022	70%
Intermediate	Ideal Indian School, Doha-Qatar	C.B.S.E	2016	75%
High School	Ideal Indian School, Doha-Qatar	C.B.S.E	2014	60%

SKILLS

Specializations:

Penetration Testing

Cybersecurity

Troubleshooting

Virtualization

Software Skills:

Canva, MS Office Suite, Oracle Virtual Box, Adobe Photoshop

WORKSHOPS:

AI Tools Workshop by Be10x.

Front End Development by DevTown.

INTERESTS/HOBBIES:

Football, Programming, Troubleshooting

PERSONAL INFORMATION

Date of Birth - 6th February, 1998

Father's Name - Aslam Saeed Ahmed Azmi

Languages Known - Arabic, Urdu, English, Hindi

Permanent Address - 140, Pathan Tola, Azmi Road, Mau, UP

DECLARATION

I hereby, declare that information provided by me is true to the best of my knowledge.

11.2. MR. AYAN AMHAD

Major:- Master of Computer Application

Email Id:- ayanay@student.iul.ac.in

Contact No.:- 7521830578

CAREER OBJECTIVE

To join a position of responsibility with a progressive organization that provides conducive learning and development.

EDUCATIONAL QUALIFICATION

Course	School/College	Board/University	Year of Passing	Marks/ Percentage
M.C.A	Integral University, Lucknow	Integral University	2024	Appearing
B.C.A	CSJM University, Kanpur	CSJM University	2022	85%
Intermediate	St. Judes School	ISC	2017	90%
High School	St. Judes School	ISC	2015	92%

SKILLS

Specializations:

Web Development

Java, Python, C/C++

Software Skills:

Canva, MS Office Suite

WORKSHOPS:

AI Tools Workshop by Be10x.

Front End Development by DevTown.

INTERESTS/HOBBIES:

Football, Programming, Reading

PERSONAL INFORMATION

Date of Birth - 28th April, 1998

Father's Name - Ikhlaq Ahmad

Languages Known - Hindi and English

Permanent Address - 269, B Awas Vikas Colony, Unnao, UP

DECLARATION

I hereby, declare that information provided by me is true to the best of my knowledge.